Wear 267 (2009) 129–135

Contents lists available at ScienceDirect

Wear

journal homepage: www.elsevier.com/locate/wear

Impact-angle dependence and estimation of erosion damage to ceramic materials caused by solid particle impact

Y.I. Oka*, S. Mihara, T. Yoshida

Department of Chemical Engineering, Hiroshima University, 1-4-1 Kagamiyama, Higashi-Hiroshima 739-8527, Japan

ARTICLE INFO

Article history:
Received 12 September 2008
Received in revised form
22 December 2008
Accepted 23 December 2008

Keywords:
Impact-angle dependence
Erosion damage
Solid particle impact
Ceramic materials
Estimation

ABSTRACT

In general, it is accepted that ceramic materials outperform metallic materials at resisting solid-particle erosion. Ceramic materials also have the commercially beneficial quality of a long life span. Nevertheless, few studies have evaluated the effectiveness of the higher erosion resistance of ceramic materials as compared with industrial metallic materials. The impact-angle dependence of erosion damage typically shows ductile and brittle manners depending on the type of material. It is very important to understand the effects of impact parameters such as velocity and particle size on impact-angle dependence of erosion damage, in order to estimate the performance of ceramic materials under given impact conditions. In this paper, predictive equations for impact-angle dependence of normalized erosion, which have been applied and verified in cases of metallic materials, were used to estimate the erosion damage to ceramic materials. In addition, the effects of impact parameters on erosion damage were also investigated. Erosion tests were conducted using a sand blast type erosion test rig. As a result, the impact-angle dependence of erosion damage to ceramic materials such as alumina, magnesia and zirconia was well characterized using exponent numbers that differed from those of metallic materials in the equation. It was found that the impact-angle dependence of erosion damage to ceramic materials shifted depending on impact parameters such as the type and size of the solid particles. The mechanisms of erosion damage also varied with the removal of either primary or secondary ceramic grains.

© 2009 Published by Elsevier B.V.

1. Introduction

Fundamental analyses on erosion caused by solid particle impact were performed in the 1960s and 1970s [1,2]. Many estimative or predictive equations have been proposed by many investigators [3], but the development of a practical and effective equation for the degradation of materials in industrial plants remains inadequate. A few researchers [4–6] have suggested properties for both ceramic materials and ceramic particles and have connected them with complicated erosion mechanisms. Estimative or predictive equations are very difficult to establish because of a wide range of impact conditions, many effective impact parameters such as velocity, angle and particle properties, and a wide variety of metallic and ceramic materials, material compositions and physical and mechanical properties. It is also difficult to estimate erosion damage to various materials at arbitrary impact angles, which is necessarily shifted by material properties, despite the importance of estimating damage from any impact-angle in an industrial setting.

We recently proposed predictive equations for erosion damage to metallic materials from any impact-angle using material hardness as a fundamental mechanical property [7–9], and applied this equation to ceramic coating materials [10]. Hardness is one of the mechanical properties of materials that is an important parameter in predictive equations and has played a primitive role in the equation of normalized erosion when estimating erosion damage at any impact-angle for candidate materials used in actual plant components. The two exponents in normalized erosion related to material hardness and constants are determined by particle properties.

The aim of this study was to confirm that the same predictive equation for normalized erosion used to investigate metallic materials could also be applied to ceramic materials, and to investigate the effect of various parameters such as type, size and velocity of particles on the impact-angle dependence of normalized erosion, thereby comparing erosion behaviours of metallic and ceramic materials. Erosion tests for ceramic materials were carried out under various particle and impact conditions. Impact-angle dependence of normalized erosion for the ceramic materials was also obtained under various impact conditions. A comparison of erosion rates between commercial ceramic materials tested in this paper and industrial metallic materials is discussed. Predictive equations

* Corresponding author.
E-mail address: kitanoyoshi49@live.jp (Y.I. Oka).

0043-1648/$ – see front matter © 2009 Published by Elsevier B.V.
doi:10.1016/j.wear.2008.12.091

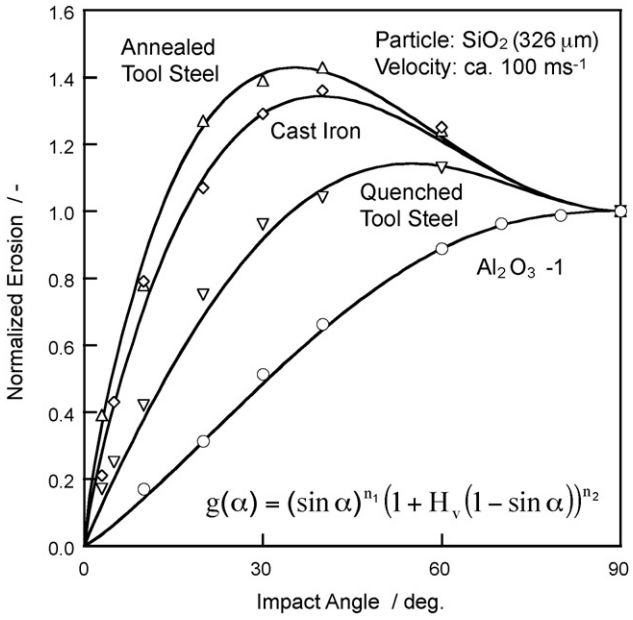


Fig. 1. Impact-angle dependence and an equation of normalized erosion for metallic and ceramic materials.

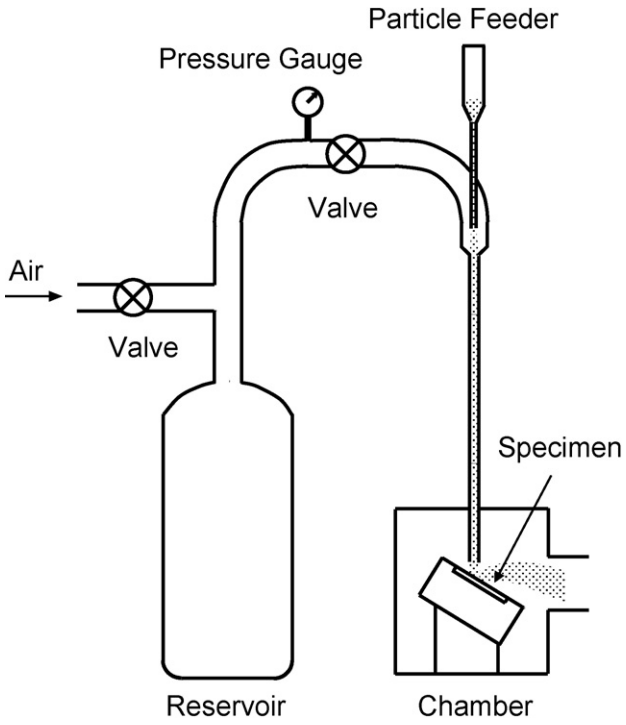


Fig. 2. Schematic diagram of a sand blast type erosion test rig.

for erosion rates at arbitrary impact angles are discussed and fundamentally proposed in this paper.

2. Equations on impact-angle dependence of erosion

Erosion damage greatly depends on the impact-angle and on the type of target material. In order to discuss the erosive effects of the impact parameters, the impact-angle must be considered. Erosion damage at arbitrary angles $E(\alpha)$ is expressed by Eq. (1), with erosion damage at 90° E_{90} [7,8].

$$E(\alpha) = g(\alpha)E_{90} \tag{1}$$

E_{90} denotes a unit of material volume per unit mass of particles (mm^3kg^{-1}). $g(\alpha)$ is the impact-angle dependence of normalized erosion with a dimentionless unit in Eq. (2).

$$g(\alpha) = (\sin\ \alpha)^{n_1}(1 + H_v(1 - \sin\alpha))^{n_2} \tag{2}$$

where H_v is the material hardness represented in units of GPa, and n_1 and n_2 are exponents determined by the target material hardness and some property (type of particle) of the particle [7,8]. The first term in Eq. (2) is associated with the brittle characteristics, or repeated fractures, of ceramic materials, compared with the repeated plastic deformation of metallic materials. The second term is associated with cutting action, which is relative and more effective at lower impact angles. $g(\alpha)$ represents the product of the two factors in terms of mutual actions. Eq. (2) has proved the impact-angle dependence of erosion damage for various metallic materials. Fig. 1 shows the normalized erosion $g(\alpha)$ of tools made of heat-treated carbon steel and cast iron, as reported in a previous paper [7]; alumina is also included, and will be shown later. The symbols are experimental data and the curves are calculation results from Eq. (2). The normalized erosion curves for every material can be perfectly expressed using only Eq. (2), although n_1 and n_2 take respective values depending on the types of materials and particles. We can estimate erosion damage at any impact-angle using Eqs. (1) and (2).

3. Experimental procedures

A sand blast type erosion test rig is shown in Fig. 2. This test rig is composed of a reservoir tank, valves, an acceleration tube and a specimen chamber. It was confirmed that the conditions of impact velocity and angle were controlled well by this test rig [11]. Impact angles were 10–90°. Impact velocities were from 39 to 128 m s^{-1}, and were controlled by a tank pressure of 0.01 to 6 bar. The impact velocity of particles differed depending on their type and size, even when the tank pressure was equal [11]. Erodent particles included angular silica sand (SiO_2) and silicon carbide (SiC) with different mean sizes of 194 or 254 to 428 μm, and round zirconia (ZrO_2) with a mean size of 200 μm, as listed in Table 1.

Materials used in this study were silicon carbide, two types of alumina (Al_2O_3-1 and Al_2O_3-2), and zirconia and magnesia (MgO) with different physical and mechanical properties, as listed in Table 1. Erosion tests by solid particle impact were conducted for several different impact angles, velocities and particles. The mass loss of a ceramic specimen per mass of erodent was measured and converted into an erosion volume rate (mm^3kg^{-1}) with each material density. Evaluations of erosion rates at an impact-angle of 90° and the impact-angle dependence of normalized erosion

Table 1
Particles and target materials used in this study.

Particle	Shape	Diameter, μm	H_v (GPa)
SiO_2	Angular	194-428	ca. 20
SiC	Angular	254-428	ca. 30
ZrO_2	Round	200	ca. 20

Material	Density (g cm^{-3})	Grain size (μm)	H_v (GPa)	F_t (MPa $m^{0.5}$)
SiC	2.9	-	23	3.6
Al_2O_3-1	3.4	10	15	3.7
ZrO_2	6.0	0.2	13	6.0
Al_2O_3-2	3.6	3	11	3.1
MgO	3.5	35	7	3.0

H_v: hardness, F_t: fracture toughness.

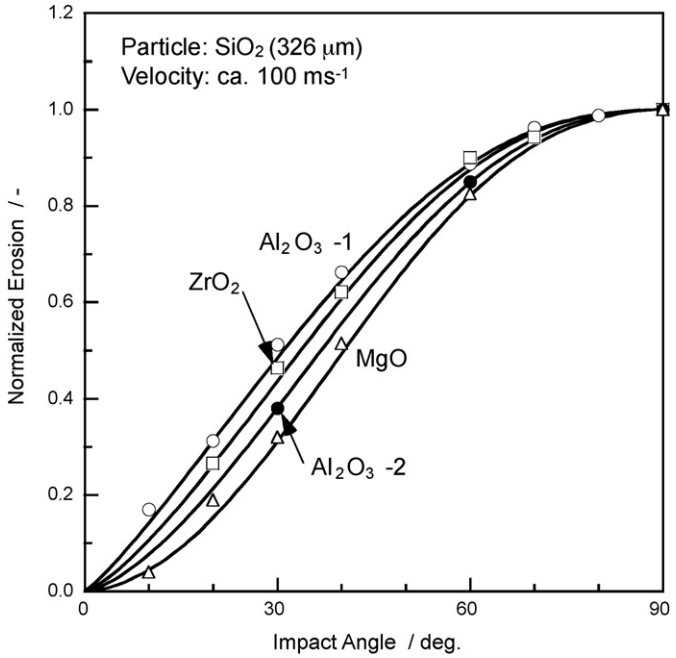


Fig. 3. Impact-angle dependence of normalized erosion for alumina (Al_2O_3-1 and Al_2O_3-2), zirconia and magnesia materials caused by silica sand particles of 326 μm.

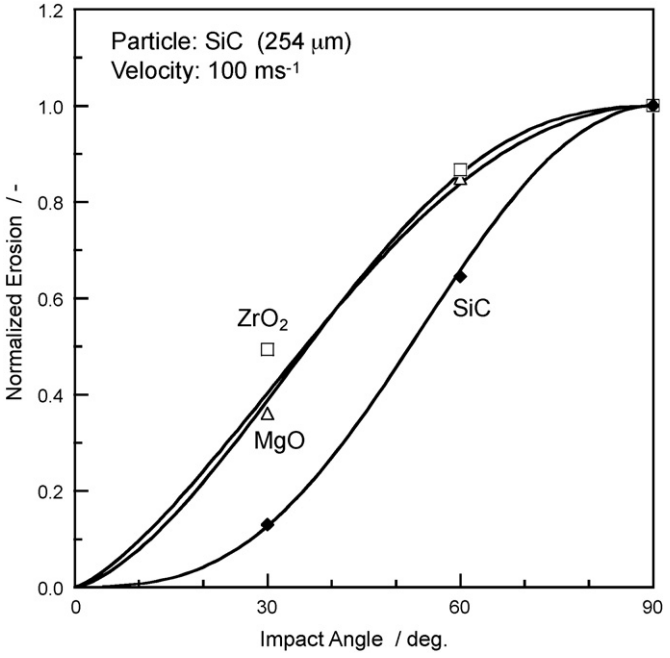


Fig. 4. Impact-angle dependence of normalized erosion for silicon carbide, zirconia and magnesia materials caused by silicon carbide particles of 254 μm.

were investigated using Eqs. (1) and (2). The curves drawn together with plotted data were based on the exponents of n_1 and n_2, as determined by respective test conditions. Erosion surfaces were observed by scanning electron microscopy (SEM) to deduce damage mechanisms and impact-angle dependence of normalized erosion for ceramic materials.

4. Results

Fig. 3 shows examples of the impact-angle dependence of normalized erosion for alumina, zirconia and magnesia at 326 μm using approximately 100 m s^{-1} of silica sand particles. Plots are experimental data and curves were drawn from Eq. (2) for each material, as determined by the respective n_1 and n_2, and by the material hardness, as shown in Table 1. It was found that the impact-angle dependence is slightly different among the target materials, but that the maximum erosion damage occurred at 90° for all materials, which is typical for ceramic materials. The curve of the harder alumina (Al_2O_3-1) shifted to a higher position, and that of the softer magnesia shifted to a lower position.

Figs. 4 and 5 show examples of impact-angle dependence for normalized erosion in cases of different particles of angular SiC (254 μm) and round ZrO_2 (200 μm) at an impact velocity of 100 m s^{-1}. The impact-angle dependence individually shifted for the types of materials and particles. The behaviour of the SiC materials damaged by SiC particles was significantly lower at lower impact angles. The impact-angle dependence of normalized erosion caused by round particles of ZrO_2 was similar irrespective of the type of material, and was lower at lower impact angles, as compared with the angular particles. Normalized erosion also reached its maximum at 90° for all ceramic materials.

Fig. 6 shows examples of the effects of particle size on the impact-angle dependence of normalized erosion for Al_2O_3-1 specimens from SiO_2 (194–428 μm) particles at an impact velocity of ca. 100 m s^{-1}. The impact-angle dependence was relatively higher depending on the gradual increase in particle size. Fig. 7 shows the effects of impact velocity on impact-angle dependence of normalized erosion for Al_2O_3-1 specimens from SiO_2 (254 μm) and SiC (326 μm) particles. The effect of impact velocity was found to be slight with the combination of an Al_2O_3-1 specimen and a SiO_2 particle. The behaviour of impact-angle dependence became high as the impact velocity increased for the two particles. The angularity and hardness number of SiC particles is higher than that of SiO_2 particles. The greater effects of impact velocity probably appeared in the case of the SiC particles.

Fig. 8 shows the SEM results of eroded surfaces of the Al_2O_3-1 specimens under impact conditions from particles of ZrO_2 (200 μm) and SiO_2 (254 and 428 μm). Many aspects of the damage from particle impact, but no morphological differences, were

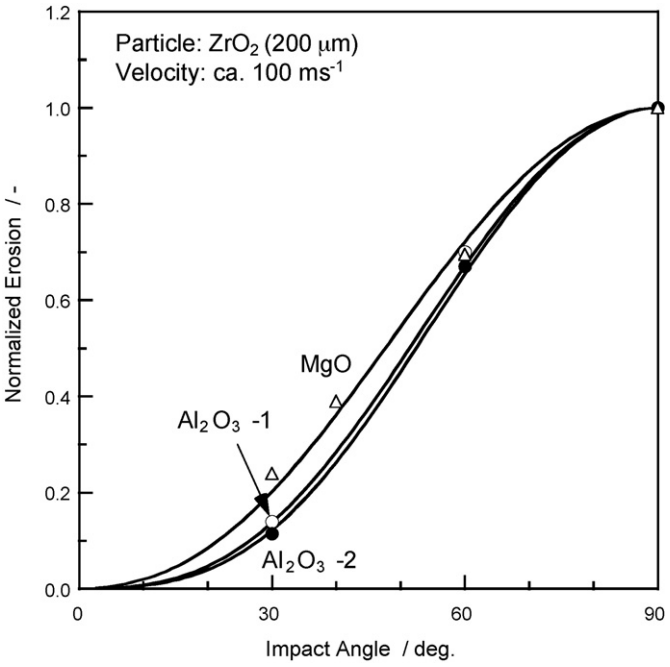


Fig. 5. Impact-angle dependence of normalized erosion for two types of alumina and magnesia materials caused by zirconia particles of 200 μm.

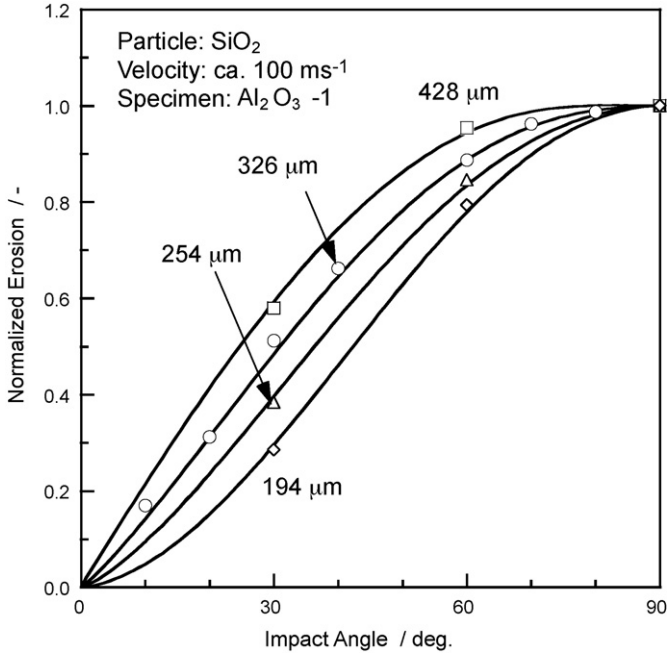


Fig. 6. Effects of particle size on impact-angle dependence of normalized erosion for alumina caused by silica sand particles.

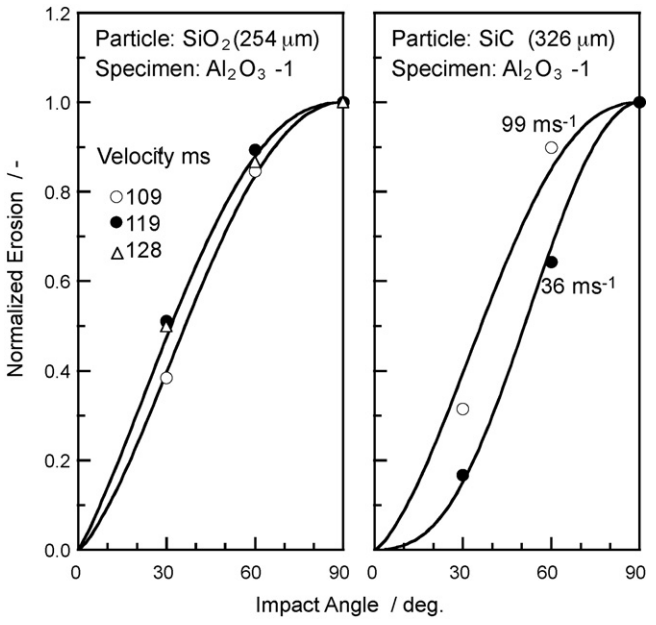


Fig. 7. Effects of particle velocity on impact-angle dependence of normalized erosion for alumina caused by silica sand and silicon carbide particles.

observed from the impact of angular and round particles at 90°. An irregular appearance (eroded and non-eroded places) could be observed on the surfaces from impact particles of 428 μm at 30°. Secondary grains, 100–200 μm larger than the primary grains (10 μm, see Table 1), were seen in the case of the alumina specimens.

Fig. 9 shows the relationships between erosion rates (mm^3kg^{-1}) at an impact-angle of 90° and at an impact velocity in a logarithmic scale. The erosion rates of magnesia were very high with all types and sizes of particles. It is evident that erosion rates caused by an angular SiC particle were greater than those caused by an angular SiO_2 particle and that erosion rates caused by a round ZrO_2 particle were unexpectedly equivalent to those caused by an angular SiO_2 particle. In general, the velocity exponent that indicates the slope of linear curves was almost equal at a range of from 1.8 to 3 [5]. The erosion rates of zirconia, silicon carbide and alumina as target materials were equivalent to those of metallic materials. The rates were obtained in the series of test results shown in Fig. 1 under the same conditions as silica sand particles.

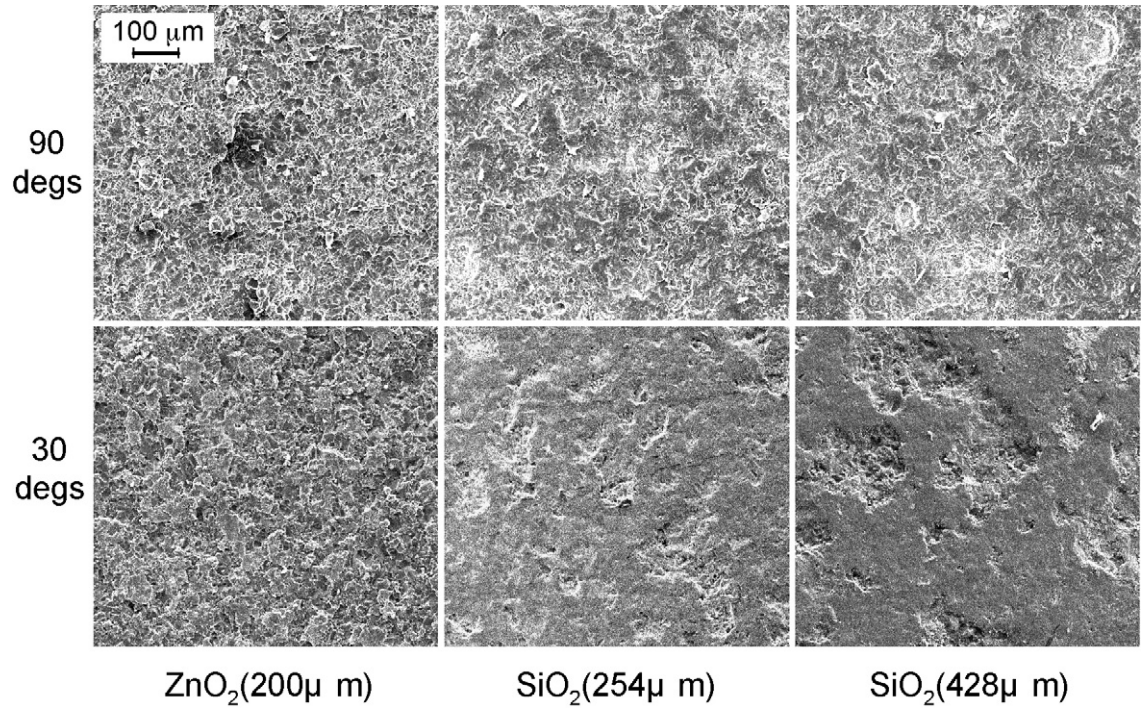


Fig. 8. SEM observations for eroded alumina specimens (Al_2O_3-1) at impact angles of 30° and 90° under conditions of different particles.

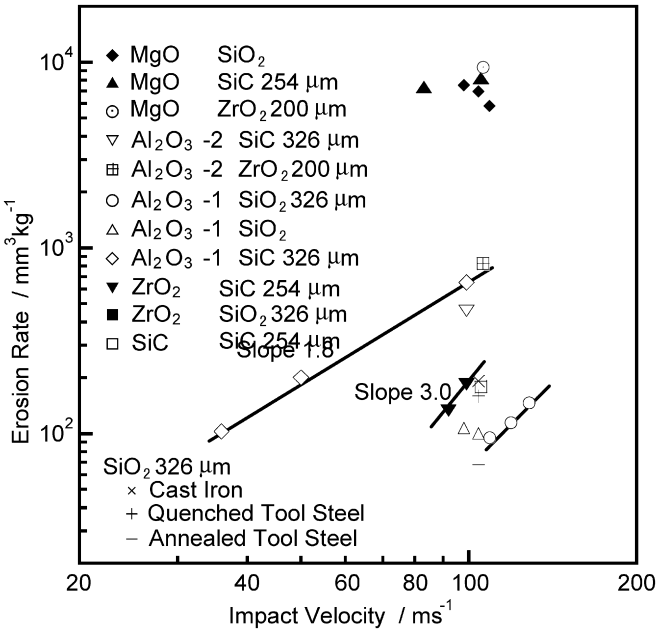


Fig. 9. Impact velocity dependence of erosion damage for various specimens and conditions.

5. Discussion

5.1. Types of ceramic materials and erosion behaviour

The erosion behaviour of a material is characterized by its erosion rate and the impact-angle dependence for normalized erosion. The erosion rates at an impact-angle of 90° are shown in Fig. 9, and those of ceramic materials were unexpectedly large despite the use of a very similar erodent as the target material. In particular, the erosion rates of ceramic materials at 90° were found to be more or less equivalent to those of carbon tool steel and cast iron under identical conditions such as velocity and type of particle [7]. Material selection from metallic to ceramic is, therefore, meaningless with respect to deeper impact angles. These results seem to be attributed to the properties and grain size of ceramic material as well as to the weak bonding between the grains (see Fig. 8). The erosion rates of magnesia were the worst among the ceramic materials used in this study. Erosion rates of ceramic materials probably depend on material supply processes, binders and sintering, which makes experimental tests an essential step in the selection of practical and useful materials.

The typical difference in impact-angle dependence between metallic and ceramic materials is shown in Fig. 1. The remarkable differences between the two types of materials appear at the lower impact angles, and the ceramic materials definitely provide high performance at lower impact angles. The performance of ceramic material is not considered to be inferior to carbon steel at lower impact angles. This would depend on the impact-angle dependence of erosion. The erosion rate was the maximum at an impact-angle of 90° for all ceramic materials. This result comes from the brittleness of ceramic materials and from the fracture mechanisms of perpendicular energy caused by particle impact.

The impact-angle dependence (the shape of curve) of erosion, or normalized erosion, is considered to be relative to the respective properties of ceramic materials (see Figs. 3–5) and to be simultaneously affected by impact conditions such as type, size and velocity of impact particles. As shown in Fig. 3, the impact-angle dependence of normalized erosion gradually shifts from up to down at low-impact angles depending on the decrease in hardness of ceramic materials for angular particles of silica sand (326 μm in diameter). However, the impact-angle dependence of normalized erosion scarcely shifted in the case of silicon carbide and zirconia particles (Figs. 4 and 5), except when both the target and particle were made of silicon carbide. Using the hardest silicon carbide for both the target and the particle decreases erosion damage at lower impact angles due to the reduction of cutting action by an angular particle. The impact-angle dependence of normalized erosion seems to be very complicated and sensitive to the hardness ratio between the target and the particle [4].

5.2. Effects of particle parameters on impact-angle dependence

The particle parameters that affect the impact-angle dependence of normalized erosion are type (shape), size and velocity. For example, the impact-angle dependence of normalized erosion for magnesia in different types of particles can be compared from Figs. 3–5, although the size of particles differs. The normalized erosion at an impact-angle of 30° was highest ($g(30) = 0.36$) for silicon carbide, median ($g(30) = 0.32$) for silica sand, and lowest ($g(30) = 0.24$) for a round zirconia particle. The angularity of a silicon carbide particle is stronger than that of a silica sand particle, and it produces a greater cutting action at lower impact angles. On the other hand, the plowing phenomenon of round particles of zirconia decreases cutting action.

The effects of particle size (194–428 μm) on impact-angle dependence of normalized erosion seemed to be larger than the other effects of particle shape and velocity. $g(30)$ for alumina shifted from 0.29 at 194 μm to 0.58 at 428 μm of a silica sand particle, as shown in Fig. 6. The alumina possessed primary small grains and secondary sintered large grains. Larger damaged areas of secondary grain size were observed at 30° for silica sand particles of 428 μm. On the other hand, a uniform damaged surface caused by the removal of primary small grains was observed for small sizes of silica sand particles, as shown in Fig. 8. Some large differences in erosion mechanisms were recognized among the different particle sizes, and were probably due to the mutual actions of relative magnifications between the sizes of target grains and the tips of particles, which are the origin of erosion damage.

The effects of impact velocity on impact-angle dependence of normalized erosion were considered to be small (Fig. 7). The reason the smaller values of normalized erosion showed at lower impact velocity was akin to the difference in the erosion mechanism caused by the mutual actions between the sizes of target grains and the tips of particles with a resultant decrease in cutting action at a low impact velocity. The effects of impact velocity were discussed for the alumina specimen (Al_2O_3-1), but it is anticipated that the effects of the other target materials depend on the size of the grains of the material relative to the particle tip configuration.

5.3. Exponents n_1 and n_2 and impact-angle dependence

As mentioned in Section 2, the n_1 and n_2 exponents reflect repeated fracture and cutting action on ceramic materials and depend on the types of target material and the impact parameters. To the contrary, the behaviour reflected by n_1 and n_2 of target materials by particle parameters suggests the effects of the type, size and velocity of the particles. Figs. 10–12 show the relationships between the respective particle parameters and n_1 and n_2 for various ceramic materials. The property of ceramic target materials is regarded here as hardness rather than the property of fracture toughness that was used in estimative equations for metallic materials [7,8].

Fig. 10 shows the effects of the types of particles and target materials on n_1 and n_2. It is obvious that the hardness of target materials is related to the n_1 and n_2 of the respective types of particles and that both positive and negative relationships will form

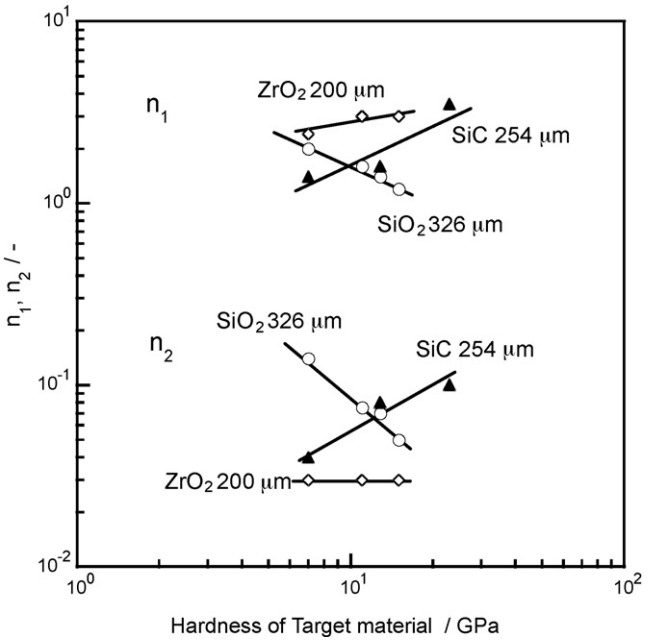


Fig. 10. Relationships between hardness of target materials and n_1 and n_2 for various ceramic materials.

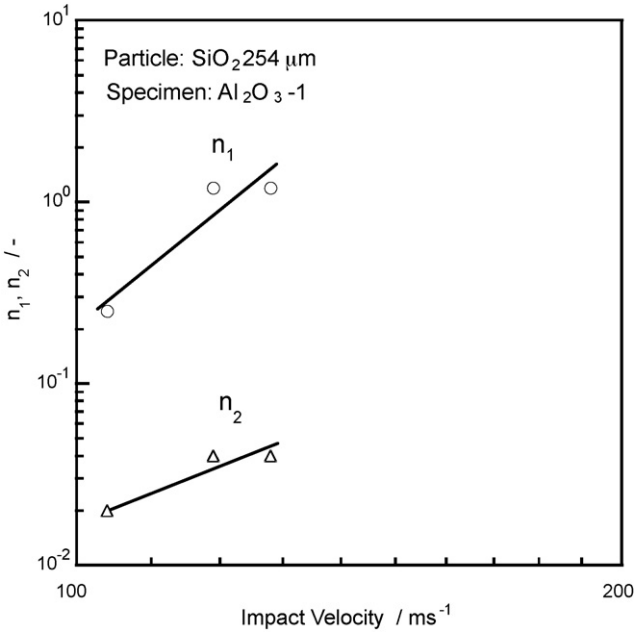


Fig. 12. Relationships between impact velocity and n_1 and n_2 for various ceramic materials.

depending on the particle type. In positive relationships of silicon carbide particles, the slopes of n_1 and n_2 were higher than those for other types of particles. Negative relationships were obtained with silica sand particles and both negative and positive relationships were obtained with zirconia particles. However, the impact-angle dependences of normalized erosion for various ceramic materials were not necessarily extreme. The relative ratio of n_1/n_2 was found to be significant in the discussion of repeated fractures and cutting action in our previous paper [12]. An n_1/n_2 of approximately 100 for round zirconia particles was the highest value among all particles, as shown in Fig. 10. This value suggests a weak cutting action and leads to a low normalized erosion $g(\alpha)$ at lower impact angles (see Fig. 5).

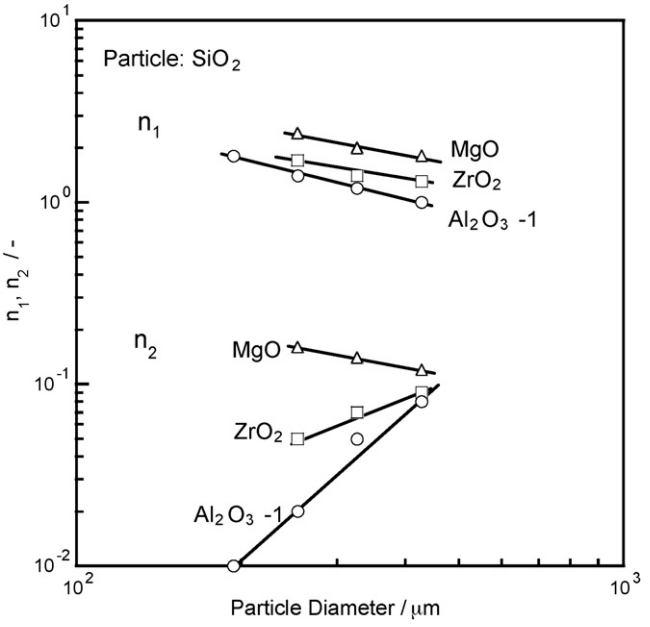


Fig. 11. Relationships between particle diameter and n_1 and n_2 for various ceramic materials.

Particle size had an effect on n_1 and n_2 for target materials (Fig. 11). The slopes of n_1 were nearly identical for the three types of ceramic materials, but the slope of n_2 was quite different. The greatest effect was in the case of the alumina (Al_2O_3-1) target material. The ratio (n_1/n_2 = 180), which was obtained for alumina (Al_2O_3-1) and silica sand particles of 194 μm, suggested a low normalized erosion $g(\alpha)$ and a low cutting action. However, a particular difference in normalized erosion was found between the high ratios of n_1/n_2 = 100 for a zirconia particle and 180 for a silica sand particle. At this point, the ratio of n_1/n_2 should be discussed for the same particle condition. The relationships between n_1 or n_2 and impact velocity for alumina and silica sand particles of 254 μm are similar within a small range of impact velocity (shown in Fig. 12), and equates to a small change in the n_1/n_2 ratio and a small effect of impact velocity on normalized erosion.

5.4. Estimation of erosion damage

The relationships between erosion rates at 90° and hardness or fracture toughness of target materials are shown in Fig. 13. Generally speaking, it is said that the erosion rate decreases with an increase in both target hardness and fracture toughness [6]. But the relationships detailed in Fig. 13 are not necessarily positive when materials having almost identical erosion rates take different values of hardness or fracture toughness. The relationships are probably connected to a combination of mechanical properties and material constituents such as primary and secondary grains and binders. Also, erosion rates exponentially increase with impact velocity (Fig. 9), so that will be estimated by an equation with exponent numbers n of impact velocity v and m of material hardness H_v or fracture toughness F_t, as written below in Eq. (3).

$$E_{90} = Kv^n(H_v, F_t)^{-m} \quad (n, m > 0) \tag{3}$$

where K is a constant that is determined by other particle parameters. Consequently, erosion rates at arbitrary impact angels $E(\alpha)$ are expressed by Eq. (4).

$$E(\alpha) = Kg(\alpha)v^n(H_v, F_t)^{-m} \quad (n, m > 0) \tag{4}$$

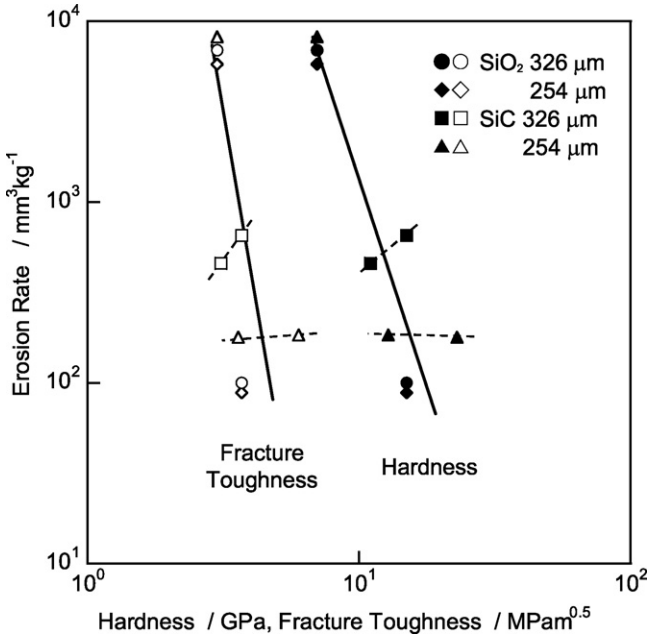


Fig. 13. Relationships between erosion rate and target hardness or fracture toughness for various ceramic materials.

The reliability of this equation depends on the relationships between the erosion rate and target hardness or fracture toughness, although normalized erosion $g(\alpha)$ is fairly good. The exponents n_1 and n_2 in normalized erosion must be estimated in the case of unknown or non-tested ceramic material. It is truly difficult, however, to estimate erosion damage, as both erosion rate and normalized erosion are definitely affected by the many impact parameters of impact velocity, type and size of particles, and probably are affected as well by the target and particle properties discussed in Sections 5.1–5.3. Based on the results of this study, when erosion resistance and normalized erosion are taken into account, the practical selection of material can be a very complicated and skillful task.

6. Conclusions

Solid particle erosion tests were conducted for various ceramic materials using particles of silica sand, silicon carbide and zirconia that were approximately ca. 100 m s^{-1}, angular and round, in order to evaluate normalized erosion and erosion rates. The conclusions are as follows:

1. Normalized erosion of ceramic materials can be calculated using the obtained exponents n_1 and n_2, according to the equation for normalized erosion as it was applied to metallic materials. The exponents n_1 and n_2 represent the effects of repeated fracture and cutting actions, respectively, and take respective values that depend on the types of target ceramic materials and particles. It is possible to estimate normalized erosion using arbitrary impact angles from 0° to 90°.
2. Normalized erosion was affected mostly by particle size and slightly by impact velocity. A change in the damage mechanism could be observed according to differences in particle size and could be attributed to the extent of damage to either primary or secondary grains of ceramic materials for alumina. The exponents n_1 and n_2 were related to target hardness, type and size of particles. The relative ratio, n_1/n_2, reflected a change in the damage mechanism.
3. The erosion rates at an impact-angle of 90° were slightly related to the hardness, or fracture toughness, of ceramic materials, but were not direct indicators of erosion damage for the various ceramic materials used in this study.
4. The erosion resistance of ceramic materials at an impact-angle of 90° was not necessarily high, but was equivalent to that of the metallic materials reported in a previous paper. At lower impact angles, however, it is possible that ceramic materials could be more effective than metallic materials at erosion resistance.

References

[1] I. Finnie, Erosion of surfaces by solid particles, Wear 3 (1960) 87.
[2] J.G.A. Bitter, A study of erosion phenomena part I, Wear 6 (1963) 5.
[3] G. Sundararajan, The solid particle erosion of metallic materials: the rationalization of the influence of material variables, Wear 186–187 (1995) 129.
[4] P.H. Shipway, I.M. Hutchings, The role of particle properties in the erosion of brittle materials, Wear 193 (1996) 105.
[5] D.W. Celotta, et al., Sand erosion testing of novel compositions of hard ceramics, Wear 263 (2007) 278.
[6] K.R. Gopi, R. Nagarajan, S.S. Rao, S. Mandal, Erosion model on alumina ceramics: a retrospection, validation and refinement, Wear 264 (2008) 211.
[7] Y.I. Oka, H. Ohnogi, T. Hosokawa, M. Matsumura, The impact angle dependence of erosion damage caused by solid particle impact, Wear 203–204 (1997) 573.
[8] Y.I. Oka, K. Okamura, T. Yoshida, Practical estimation of erosion damage caused by solid particle impact. Part 1. Effects of impact parameters on a predictive equation, Wear 259 (2005) 95.
[9] Y.I. Oka, T. Yoshida, Practical estimation of erosion damage caused by solid particle impact. Part 2. Mechanical properties of material directly associated with erosion damage, Wear 259 (2005) 102.
[10] Y.I. Oka, T. Yoshida, Y. Yamada, T. Yasui, S. Hata, Evaluation of erosion and fatigue resistance of iron plated chromium nitride applied to turbine blades, Wear 263 (2007) 379.
[11] Y.I. Oka, N. Nishimura, K. Nagahashi, M. Matsumura, Control and evaluation of particle impact conditions in a sand erosion test facility, Wear 250 (2001) 736.
[12] Y.I. Oka, M. Nisimura, K. Nagahasi, M. Matumura, Impact angle dependence of erosion by solid particle impact for metallic materials, Zairyo-to-Kankyo (JSCE) 48 (1999) 355.